



PROJECT PROPOSAL

SUBMITTED TO: MISS NOREEN KHALID

ClassSync AI

BSCS

v1.0

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BACKGROUND AND JUSTIFICATION

University course scheduling remains a complex challenge due to overlapping classes, instructor availability, and resource constraints.

- University course scheduling is complex, involving student enrollments, instructor availability, and time slot constraints.
- Traditional scheduling methods are inefficient, leading to conflicts and inconsistencies.
- AI and ML offer a data-driven solution, improving automation, fairness, and scalability.
- An adaptive AI system can optimize timetables, balancing course loads and enhancing resource utilization.



Problem Statement

- University scheduling faces major challenges due to:
 - Overlapping courses, instructor constraints, and limited classrooms.
 - Manual and rule-based scheduling, leading to inefficiencies, conflicts, and student-instructor dissatisfaction.
 - Lack of adaptability to real-time changes, including demand fluctuations and last-minute adjustments.
- This research aims to develop an AI-driven system that:
 - Uses historical data and student enrollment trends to optimize scheduling.
 - Reduces conflicts, balances course loads, and improves efficiency.
 - Enhances faculty workload management and student learning experiences.



Aims and Objectives

- To develop and validate an AI-driven scheduling system that:
 - Optimizes university timetables with minimal conflicts.
 - Enhances scheduling efficiency through automation.
 - Adapts dynamically to real-time constraints.
 - Ensures fair resource allocation for students and faculty.



PREVIOUS RESEARCH GAP

- Existing scheduling methods rely on manual, rule-based, and heuristic approaches, leading to inefficiencies.
- AI-driven scheduling research has focused on industries and workforce management, with limited implementation in universities.
- Existing AI-based solutions lack real-world university testing, making them less adaptable to academic constraints.
- Current methods fail to accommodate real-time changes, fairness in class distribution, and student preferences.
- There is a need for an AI-driven approach that optimizes course allocation, reduces conflicts, and improves scheduling efficiency.

Objectives

- Develop an AI-driven scheduling system that integrates course schedules, faculty availability, student enrollments, and classroom capacities.
- Ensure real-time adaptability by handling course demand shifts, faculty unavailability, and last-minute changes.
- Implement machine learning algorithms for conflict detection, resource optimization, and predictive scheduling.
- Improve student and faculty experience by ensuring conflict-free schedules, fair workload distribution, and balanced class assignments.
- Provide interactive visualization tools for real-time analysis of schedules, course distribution, and resource utilization.
- Enable user feedback integration to refine and optimize the system for better usability and scalability.

Research Questions

- How can AI-driven algorithms enhance the accuracy and efficiency of university timetable generation while considering course availability, instructor preferences, and classroom constraints?
- What impact does a data-driven scheduling system have on reducing conflicts, balancing faculty workload, and improving student satisfaction compared to traditional scheduling methods?
- Can machine learning models predict and adapt to changing scheduling constraints, such as faculty availability, student demand, and classroom occupancy, to ensure an optimized and scalable timetable system?



Significance of Research

The AI-driven scheduling system minimizes scheduling conflicts and errors by optimizing faculty allocations, reducing classroom clashes, and improving resource utilization. It enhances student and faculty satisfaction by eliminating long gaps, preventing course overlaps, and ensuring fair workload distribution. As a scalable and adaptable solution, it can adjust to different university structures, faculty preferences, and semester variations. Additionally, it improves academic planning through data-driven insights, making scheduling more efficient and informed. By enabling AI-driven adaptability, the system future-proofs university timetables, allowing institutions to handle last-minute schedule changes without requiring manual rework.

RESEARCH METHODOLOGY

PHASE 01

DATA COLLECTION & PREPROCESSING

- **DATA SOURCES**

- **STUDENT ENROLLMENT DATA**
- **INSTRUCTOR AVAILABILITY & CONSTRAINTS**
- **CLASSROOM SCHEDULES & CAPACITIES**
- **COURSE DEPENDENCIES (PRE-REQUISITES, LAB-LECTURE PAIRING)**

- **PREPROCESSING STEPS**

- **CLEANING: REMOVE MISSING/DUPLICATE ENTRIES**
- **NORMALIZATION: STANDARDIZE FORMATS ACROSS INSTITUTIONS**
- **SEGMENTATION: SPLIT INTO TRAINING, VALIDATION, AND TESTING SETS**



KAGGLE DATASET

 student_id 	# year_semester 	 course_id 	 course 	 weekday 	 start_time 	 end_time 	 room_address 
Anonymous id for an undergraduate student	Year (2019) and semester (1 or 2)	Id that identifies a certain course	Name of the course	Weekday when the class takes place	Time when the class starts	Time when the class ends	Adress of the room where the class takes place
0 total values	610934 total values	621910 total values	[null] 2 100% 0%	[null] 3/FSD 100% 0%	622599 total values	622599 total values	[null] 100%
100115	20191	192015	Primary Teaching	Monday	08:00	11:40	FE-5 - SALA DE AULA 04
100115	20192	142166	Supervised Internship in French 1	Tuesday	16:00	17:50	Missing info
100115	20192	142166	Supervised Internship in French 1	Thursday	16:00	17:50	Missing info
100140	20191	139190	Social History And General Policy	Tuesday	16:00	17:50	SALA PAT AT 045
100140	20191	134911	Political Sociology	Wednesday	14:00	15:50	SALA PAT AT 061
100140	20191	139190	Social History And General Policy	Thursday	16:00	17:50	SALA PAT AT 045
100140	20191	134911	Political Sociology	Monday	14:00	15:50	SALA PAT AT 061
100140	20192	134635	Class Structure And Social Stratification	Thursday	14:00	15:50	SALA PJC BT 012
100140	20192	134635	Class Structure And Social Stratification	Tuesday	14:00	15:50	SALA PJC BT 012

PHASE 02

MODEL DEVELOPMENT & VALIDATION

- **MACHINE LEARNING MODELS USED:**

- **CONSTRAINT SATISFACTION MODEL → ENSURES FEASIBILITY**
- **GENETIC ALGORITHM → OPTIMIZES SCHEDULING**
- **REINFORCEMENT LEARNING → ADAPTS IN REAL-TIME**

- **VALIDATION METRICS:**

- **SCHEDULING ACCURACY (CONFLICT-FREE SCHEDULES)**
- **CONFLICT RESOLUTION RATE (HANDLING OVERLAPS)**
- **OPTIMIZATION EFFICIENCY (TIME TAKEN TO GENERATE SCHEDULES)**



PHASE 03

DEPLOYMENT & EVALUATION

- **IMPLEMENTATION**

- **AUTOMATED SCHEDULE GENERATION**
- **INSTRUCTOR & STUDENT PREFERENCE INTEGRATION**
- **CUSTOMIZABLE DASHBOARD FOR ADJUSTMENTS**

- **EVALUATION:**

- **COMPARE AI-GENERATED VS. MANUAL SCHEDULES**
- **FACULTY & STUDENT FEEDBACK COLLECTION**
- **PERFORMANCE REFINEMENT BASED ON REAL-WORLD TESTING**

AI MODEL

Model	Purpose	Strengths
Constraint Model	Feasibility Check	Prevents conflicts
Genetic Algorithm	Optimization	Improves efficiency
Reinforcement Learning	Adaptive Scheduling	Handles real-time changes

COMPARISON

Importance

Provides a scalable model for AI-driven academic management.

Enhances efficiency and automation in university scheduling.

Reduces manual workload and scheduling conflicts.

Supports fair and optimal resource allocation.

CONCLUSION

Purpose of Research:

Develop an AI-driven scheduling system to optimize university timetables and minimize conflicts.

16

Key benefits:

- Automates scheduling for efficiency and fairness.
- Provides real-time adaptability to dynamic constraints.
- Reduces administrative workload and improves resource utilization.

Enhances university scheduling, ensuring a scalable and intelligent approach to academic management.

Thank you